

ZHEZHOU LI 李哲舟

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EDUCATION

Lanzhou University

Bachelor of Science in Physics

Lanzhou, China

Sep. 2023 – Jun. 2027 (Expected)

- GPA: 3.23/5.00
- Key Coursework: Foundations of Computational Physics (incl. experiments) (**94/100**); Python and Data Science Practice (**97/100**).

University of California, Berkeley

Summer Session Visiting Student

Berkeley, USA

Jun. 2024 – Aug. 2024

- GPA: 4.2/5.0 (High Distinction)
- Coursework: Digital Infrastructure, examining the physical foundations of global communication networks and AI-scale computation.

RESEARCH INTERESTS

Focus Areas: AI for Science (AI4S), Complex Systems, AI Agents for Scientific Workflows, Collective Behavior, Nonlinear Dynamics, Neuroscience, Cellular Automata.

ONGOING RESEARCH PROJECTS

Meteor-Radar Signal Processing, MacDM-like Methods, and AI-native Research Workflow

May 2026 – Present

Research Assistant

Key Laboratory of New Generation Artificial Intelligence Technology and Its Interdisciplinary Applications (Southeast University), Ministry of Education, China (Prof. Shipeng Zhu)

- **Project Initiation & Partnership Building:** Initiated the research idea, proactively connected SEU and CETC-22 stakeholders, aligned research needs and data-access expectations, and helped establish the active collaboration.
- **MacDM-like Radar Methodology:** Exploring a readiness pipeline for meteor radar searches motivated by Macroscopic Dark Matter (MacDM), emphasizing raw signal processing, physics-informed anomaly scoring, validation design, and conservative claim boundaries.
- **AI-native Research Workflow:** Designing agent-assisted workflows for station operations and radar-data analysis, covering evidence-chain diagnostics, MCP-style tool governance, human review, and auditable experiment records.
- **PINN Wind-Field Inference:** Investigating Physics-Informed Neural Network (PINN) approaches for inferring upper-atmospheric wind fields from meteor-trail radar observations and related physical constraints.

Forest Cellular Automaton for Tree-Size Structure, Disturbance Cascades, and Functional Loss

Jan. 2026 – Present

Research Assistant

Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences (Prof. Kailiang Yu)

- **SOC/HOT Theory Grounding:** Conducted early-stage literature synthesis on Self-Organized Criticality (SOC) and Highly Optimized Tolerance (HOT), broadening the theoretical basis for subsequent disturbance-cascade modeling.
- **Data-Constrained Forest CA:** Developing a forest cellular automaton linking allometric tree-size states, local neighborhood structure, Motter-Lai load-capacity cascades, and biomass-loss proxies.
- **Null-Model Validation:** Designed null-model, permutation-test, and distribution-fitting workflows to test mechanism associations and flag unsupported power-law or fractal-emergence claims.

RESEARCH EXPERIENCE

Algorithmic Criticality in Turbulence: Data-Driven Detection of Roughness-Induced Phase Transitions via Parallel Symbolic Enumeration Oct. 2025 – Jul. 2026

Research Assistant

Center for Statistical Physics and Complex Systems, Institute of Theoretical Physics, LZU (Prof. Zhixi Wu)

- **PSE-Based Turbulence Analysis:** Designed a white-box AI4S workflow combining Parallel Symbolic Enumeration, sliding-window symbolic regression, entropy diagnostics, and high-Reynolds-number dataset curation ($Re: 10^3 \sim 10^8$) to map friction-law transitions across turbulence regimes.

Geometric Deep Learning for Complex System Resilience & Property Prediction Feb. 2025 – May 2026

Research Assistant

Center for Statistical Physics and Complex Systems, Institute of Theoretical Physics, LZU (Prof. Zhixi Wu)

- **HGCN Resilience Modeling:** Implemented Hyperbolic Graph Convolutional Networks for complex-system resilience and property prediction, while developing resilience-quantification workflows and surveying topology-dynamics inference bottlenecks.

Independent Research Software for Agentic Scientific Workflows May 2026 – Present

Independent Developer

- Built **agent-ready**, a Node.js toolkit for repository-level agent contracts, cross-agent adapters, deterministic evals, evidence gates, and claim-boundary reports.
- Developed a React and TypeScript complex-systems visualization harness with Canvas 2D and WebGPU WGSL verifiers for auditable, language-assisted control of nonlinear generative models.

PUBLICATIONS & MANUSCRIPTS

- [1] **Zhezhou Li**, et al. “Automated Discovery and Phase Characterization of Turbulence Friction Scaling Laws via Parallel Symbolic Enumeration.” *Manuscript in preparation, targeting Physical Review E*.
- [2] **Zhezhou Li** and Zhixi Wu. “A Hyperbolic Graph Convolutional Network (HGCN) Framework for Resilience Quantification in Complex Systems.” *Submitted manuscript, under review*.
- [3] **Zhezhou Li** and Kailiang Yu. “Forest Cellular Automaton for Tree-Size Structure, Disturbance Cascades, and Functional Loss.” *Preprint in preparation, targeting PNAS*.

SKILLS

- **Programming:** Python (PyTorch, NumPy, Pandas, Pygame, NetworkX), C, MATLAB, LaTeX; experienced with AI-assisted coding workflows.
- **AI Agent Harness & Research Workflow:** Agent harness configuration, AGENTS.md/project contracts, context engineering, tool-use/MCP governance, subagent coordination, and evidence gates.
- **Experimental Techniques:** Modern Physics, Fundamental Mechanics, Electricity, and Optics experiments; experimental data processing and theoretical analysis.
- **Languages:** English (TOEFL 96), Mandarin (Native).

LEADERSHIP & COMMUNITY ENGAGEMENT

Lanzhou University Official Media Center

Deputy Director of Photography Department

Jun. 2025 – Present

- Coordinate photography-team workflows and visual coverage for university events.

Philosophy and Dialectics Society (High School Alumni)

Founder & Strategic Advisor

Jun. 2023 – Present

- Founded the society and advise event development; helped manage a “Mentor-Mentee” program facilitating hundreds of matches.